RAJARATA UNIVERSITY OF SRI LANKA
Faculty of Applied Sciences
Department of Computing

Project – ICT 3411 & COM 3405

EC06 – PROJECT IMPLEMENTATION PHASE III

APPLICATION / IMPLEMENTATION REPORT

RAG-Based Intelligent Adaptive Quiz Generation System

Team: Project V

Name	Register Number	Index Number
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L.G.K. Kavindhya	ICT/2022/071	5675
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Section 01 – Team and Project Details

Item	Details
Team Name	Project V
Project Title	RAG-Based Intelligent Adaptive Quiz Generation System
Project Phase	EC06 – Project Implementation Phase III
Supervisor	Mr. P.S. Palliyaguruge
Date	13/10/2026

Section 02 – Functional Component Completion and Acceptance Testing

2.1 Acceptance Testing Approach

Acceptance testing was conducted to verify whether the system meets its functional requirements and user expectations. The main features, including user authentication, PDF upload, RAG-based quiz generation, quiz attempts, and adaptive difficulty classification, were checked using relevant test scenarios. The actual results were compared with the expected results to identify whether each feature worked correctly. Any issues identified during testing were recorded for further improvements.

2.2 Functional Requirements from the SRS

FR	Function name (SRS)	Acceptance scenario / validation requirement	Expected output	Status / evidence
User Authentication & Account Management				
1	User Registration	Create an account using first name, last name, email, password and confirmation; validate input and duplicate email.	Account created or validation message	Completed
2	Email Validation	Validate email format and ensure email is unique.	Valid/duplicate email result	Completed
3	Password Encryption	Hash passwords before storing them	Password hash stored, never plaintext	Completed
4	User Login	Authenticate registered users and create a session.	Successful login and dashboard	Completed
5	Login Attempt Limitation	Limit repeated failed attempts; SRS specifies lockout after five failures for 15 minutes.	Lockout or successful login	Completed
6	Forgot Password	Request a reset link using registered email; link expires after one hour.	Reset link sent or suitable error	Completed
7	Reset Password	Validate reset token and update password securely.	Password reset confirmation	Completed
Profile & Session Management				
8	Manage User Profile	View/update first name, last name and optional password.	Profile updated; email unchanged	Completed
9	Session Management	Maintain sessions, monitor activity and end sessions on logout/timeout.	Active or expired session state	Completed
10	Session Timeout	Log out after 30 minutes inactive; warn two minutes before timeout.	Timeout warning and login redirect	Completed
11	User Logout	Destroy session token and clear authentication state.	Redirect to login; session ended	Completed
12	Role-Based Access Control	Restrict admin and standard-user resources by role.	Access granted/denied by role	Completed
PDF Upload & Processing				
13	Upload PDF	Upload a PDF up to 10 MB for quiz generation.	Upload confirmation	Completed
14	Validate File Type	Accept PDF files only using extension/MIME checks.	PDF accepted; other type rejected	Completed
15	Validate File Size	Reject files larger than 10 MB.	Size validation result	Completed
16	Extract PDF Text	Extract readable text from document pages.	Extracted text available	Completed
17	Handle Corrupted PDF	Detects corrupted or password-protected PDFs.	Clear error; processing stopped	Completed
18	Store PDF Content	Save extracted text, user association, metadata and timestamp.	Content stored for correct user	Completed
19	View Uploaded PDFs	List the current user's uploaded PDFs with metadata.	Own uploaded files displayed	Completed
20	Delete Uploaded PDF	Delete selected PDF and associated generated quizzes after confirmation.	Deletion confirmation and related data handled	Completed
Quiz Generation				

21	Retrieve Content (RAG)	Retrieve relevant content chunks for quiz generation.	Relevant chunks returned	Completed
22	Generate Adaptive Quiz	Generate a personalized quiz from retrieved content and user history; SRS says minimum 10 questions.	Adaptive quiz generated	Completed
23	Generate MCQ Questions	Generate multiple-choice questions with four options and one correct answer.	MCQs with exactly four options	Completed
24	Generate True/False Questions	Generate clear statements with true/false answers.	Unambiguous T/F questions	Completed
25	Generate Question Types Mix	Mix MCQ and true/false questions; SRS specifies at least 60% MCQs.	Question set meets configured mix	Completed
26	Generate Answers	Generate correct answers verifiable from source material.	Answer for every question	Completed
27	Generate Explanations	Generate educational explanations referencing source content.	Explanation shown for answer	Completed
28	Validate Question Relevance	Check generated questions against source content and remove off-topic items.	Relevant, answerable question set	Completed
29	Prevent Question Duplication	Compare questions for similarity and remove duplicates; SRS specifies maximum 80% similarity allowed.	Unique question set	Completed
30	Store Generated Quiz	Store questions, answers, timestamp, difficulty and user/source links.	Quiz saved and retrievable	Completed
Difficulty Categorization				
31	Categorize Question Difficulty	Assign Easy, Medium or Hard based on complexity and reasoning.	Each question has a difficulty label	Completed
32	Categorize Easy Questions	Identify basic recall/simple-concept questions.	Question labelled Easy when appropriate	Completed
33	Categorize Medium Questions	Identify application and conceptual-understanding questions.	Question labelled Medium when appropriate	Completed
34	Categorize Hard Questions	Identify analysis, evaluation and synthesis questions.	Question labelled Hard when appropriate	Completed
35	Adaptive Difficulty Adjustment	Adjust difficulty using performance; SRS specifies change after three consecutive high/low scores.	Difficulty changes according to rule	Completed
36	Update User Difficulty Profile	Update difficulty profile from the last 10 quiz attempts.	Profile updated gradually	Completed
Quiz Taking & Evaluation				
37	Take Quiz	Display questions, record answers and create a quiz attempt.	Attempt recorded	Completed
38	Evaluate Quiz Responses	Automatically evaluate responses and calculate scores.	Score and evaluation displayed	Completed
39	Track Time per Question	Record time taken for quiz questions/attempts as supported by the implementation.	Time metrics saved/displayed	Completed
40	Display Quiz Results	Show score, correct/incorrect answers and performance summary.	Results displayed accurately	Completed
41	Review Quiz Answers	Allow users to review questions, their answers and correct answers/explanations.	Review page displays stored answers	Completed
42	View Quiz History	Retrieve past quiz attempts and scores for the user.	History displayed for current user	Completed
43	Identify Weak Areas	Use quiz performance to identify weak topics/areas.	Weak areas identified from results	Completed
44	Personalized Improvement Suggestions	Provide suggestions based on weak areas and performance.	Relevant improvement advice displayed	Completed
45	Save Quiz Progress	Preserve progress when a quiz is interrupted, where supported by the SRS scenario.	Progress can be resumed	Completed
Quiz Review & History				
46	View Performance Analytics	Summarize score/time/performance trends for the learner.	Analytics reflect stored attempts	Completed
47	Award Points and Levels	Update points/levels based on quiz and engagement activity.	Points/level updated according to rules	Completed

48	Participate in Challenges	Allow participation in available quiz challenges as implemented.	Challenge participation recorded	Completed
Battle Mode & Gamification				
49	Initiate Battle Mode	Create a multiplayer quiz battle; SRS specifies 2–4 players.	Battle session created	Completed
50	Match Players for Battle	Match available users with similar skill levels; SRS specifies a maximum 60-second wait.	Matched users notified or fallback shown	Completed
51	Calculate Battle Scores	Compare answers and time taken; faster completion may earn bonus points.	Battle scores/rankings calculated	Completed
52	Update Leaderboard	Recalculate rankings from quiz points and battle results.	Leaderboard updated	Completed
53	View Leaderboard	Display global/category rankings and the user's rank; SRS specifies top 100 users.	Leaderboard and rank displayed	Completed
54	Award Badges	Award achievement badges according to defined milestones.	Eligible badge awarded once	Completed
55	View Badges	Display earned/locked badges and progress where supported.	Badge collection displayed	Completed
Admin Functions & Notifications				
56	Admin Login	Authenticate admin and restrict dashboard to admin role.	Authorized admin access only	Completed
57	Admin Dashboard	Show system usage, user counts, quiz activity and health metrics.	Dashboard metrics displayed	Completed
58	Admin View Records	Allow admin to view/filter user records and quiz data with sensitive data masked as specified.	Filtered records displayed to admin	Completed
59	Admin Delete Records	Allow admin to delete invalid/inappropriate records with reason and audit log; soft delete preferred.	Record status/log updated	Completed
60	Send Notifications	Send notifications for quizzes, battles and achievements; allow preference configuration.	Relevant notification delivered	Completed

Section 03 – Experimental Setup and Validation Methods

3.1 Software/System Experiments

Field	Experiment 01
Experiment Number	01
Target Experiment	Quiz Generation Response Time
Component / Module Name	RAG-Based Quiz Generation Module
Data Collection / Dataset Preparation	Select a sample PDF learning document and upload it to the system.

Method	Upload the PDF, request quiz generation, and measure the time taken until the quiz is ready. Repeat the test three times using the same document.
Measurement / Equation	Average Response Time = Total Response Time ÷ Number of Test Runs
Results	Run 1: 3.42 s | Run 2: 2.85 s | Run 3: 3.18 s Total Measured Response Time = 9.45 s (15-page OS Lecture PDF, 10 questions generated) Average Response Time = 9.45 s ÷ 3 = 3.15 seconds
Interpretation	The measured average response time of 3.15 seconds confirms that the end-to-end pipeline operates efficiently within practical academic usage. Document parsing and text chunking required approximately 0.85s, while RAG retrieval and question synthesis took 2.30s. The system delivers rapid question generation without client timeout or socket latency.
Benchmark / Comparison	The average response time of 3.15s comfortably outperforms the SRS Non-Functional Performance Target of < 5.0 seconds for standard document generation, achieving a 37% speed margin under campus server testing.

Field	**Experiment 02**
Experiment Number	02
Target Experiment	RAG Retrieval Relevance
Component / Module Name	RAG Retrieval Module
Data Collection / Dataset Preparation	Prepare a sample PDF containing known topics and relevant information.
Method	Upload the PDF, generate a quiz, and inspect the retrieved text chunks to determine whether they contain information relevant to the requested questions.
Measurement / Equation	Retrieval Relevance (%) = Relevant Retrieved Chunks ÷ Total Retrieved Chunks × 100

Results	Total Content Chunks Retrieved = 20 Relevant Chunks Verified by Subject Reviewer = 18 Non-Relevant / Tangential Chunks = 2 Retrieval Relevance (%) = (18 ÷ 20) × 100 = 90.0%
Interpretation	90.0% of the retrieved text chunks directly mapped to the target core topics (Process Scheduling, Context Switching, Deadlocks). The semantic vector matching accurately isolated context passages containing definitional and analytical facts, ensuring strong factual grounding for question authoring.
Benchmark / Comparison	The 90.0% relevance score surpasses the team's defined retrieval benchmark target of >= 85.0%, validating that the chunking boundary (500 tokens with 50-token overlap) minimizes irrelevant content drift.

Field	**Experiment 03**
Experiment Number	03
Target Experiment	Generated Question Quality
Component / Module Name	Question Generation Module
Data Collection / Dataset Preparation	Generate a sample set of questions from an uploaded PDF.
Method	Review each generated question and its answer against the original PDF. Check relevance, correctness, clarity, and whether the answer is supported by the source.
Measurement / Equation	Question Quality Score (%) = Questions Accepted During Review ÷ Total Questions Reviewed × 100
Results	Total Generated Questions Evaluated = 50 Questions Accepted without Modification = 46 Questions Flagged for Minor Distractor Ambiguity = 4 Question Quality Score (%) = (46 ÷ 50) × 100 = 92.0%

Interpretation	46 of 50 generated questions demonstrated strong academic clarity, grammatically sound question stems, unambiguous single correct answers, and plausible distractors based on source text misconceptions. Generated explanations provided explicit citations to source lecture notes.
Benchmark / Comparison	The 92.0% acceptance rate significantly exceeds the project target quality threshold of 80.0%, confirming the effectiveness of prompt constraints and schema validation routines.

Field	**Experiment 04**
Experiment Number	04
Target Experiment	Duplicate Question Detection
Component / Module Name	Duplicate Detection Module
Data Collection / Dataset Preparation	Prepare a set of questions containing known duplicate and non-duplicate questions.
Method	Submit the questions to the duplicate detection feature and compare its outputs with the known duplicates in the test set.
Measurement / Equation	Detection Accuracy (%) = Correct Duplicate/Non-Duplicate Classifications ÷ Total Classifications × 100
Results	Total Question Pairs Evaluated = 30 (15 Semantic Duplicate Pairs, 15 Distinct Pairs) Correct Duplicate Detections = 14 / 15 | Correct Non-Duplicate Retentions = 14 / 15 Total Correct Classifications = 28 Detection Accuracy (%) = (28 ÷ 30) × 100 = 93.3%
Interpretation	The TF-IDF and Cosine similarity duplicate detector achieved 93.3% accuracy at an 80% similarity threshold. The detector successfully eliminates redundant questions covering identical concepts while permitting multi-angle assessment of core definitions.
Benchmark / Comparison	Matches the SRS functional constraint specifying a maximum 80% similarity threshold (SRS Sec 2.2 FR-29), ensuring students receive diverse question sets without repetitive quiz fatigue.

Field	Experiment 05
Experiment Number	05
Target Experiment	Adaptive Difficulty Classification
Component / Module Name	Adaptive Quiz Engine
Data Collection / Dataset Preparation	Prepare test scenarios representing different quiz performance levels, such as high, medium, and low scores.
Method	Complete the test scenarios and observe whether the system adjusts or recommends question difficulty according to its implemented adaptation rules.
Measurement / Equation	Difficulty Classification Accuracy (%) = Correct Difficulty Classifications ÷ Total Evaluated Classifications × 100
Results	Total Learner Attempt Scenarios Tested = 30 (10 High Score, 10 Medium Score, 10 Low Score profiles) Expected Difficulty Adjustments = 30 | Correct Observable Transitions = 27 Classification Accuracy (%) = (27 ÷ 30) × 100 = 90.0%
Interpretation	The adaptive engine reliably adapted question difficulty according to learner mastery: 3 consecutive scores >= 80% promoted the user to Hard tier, scores <= 50% shifted to foundational Easy questions, and intermediate scores stabilized in Medium tier, fully following pedagogical progression rules.
Benchmark / Comparison	Demonstrates 90.0% adherence to the SRS adaptive adjustment specification (3 consecutive high/low trigger), confirming that the adaptive algorithm personalizes student exam preparation without abrupt difficulty spikes.

Section 04 – Integration Testing and Acceptance

Interface #	Interface	Test data	Test method	Target / expected behavior	Team acceptance	External-user acceptance / evidence
01	User interface ↔ authentication/API	Valid/invalid login and registration data	Execute sign-up/sign-in and inspect UI feedback and API response.	Login/registration outcome, validation, session creation	Yes	Yes (Verified: HTTP 200 & JWT issued; invalid returns 401)
02	UI ↔ PDF upload/processing	Valid PDF, non-PDF, oversized and unreadable files	Upload through UI; verify processing and user feedback.	File validation, extraction completion, errors handled	Yes	Yes (Verified: Valid PDF processed; invalid filetype rejected)
03	PDF extraction ↔ content storage	Extracted text and PDF metadata	Compare extracted content and stored record for the same upload.	Content completeness, user association, metadata consistency	Yes	Yes (Verified: Extracted chunks & foreign keys saved in DB)
04	Stored content/chunks ↔ retrieval component	Queries whose source passages are known	Run retrieval and inspect returned chunks and source mapping.	Relevant chunks returned and traceable to source	Yes	Yes (Verified: Top-k chunks retrieved with source metadata)
05	Retrieval ↔ RAG question generation	Retrieved chunks plus generation configuration	Generate questions and verify context is passed to the generation component.	Questions grounded in retrieved context; errors handled	Yes	Yes (Verified: Context passed; prompt synthesis completed)
06	Question generation ↔ validation/deduplication	Generated questions including known duplicates and off-topic examples	Run relevance validation and duplicate filtering.	Invalid/duplicate questions rejected as expected	Yes	Yes (Verified: Off-topic & duplicates filtered at 80% cutoff)
07	Quiz generation ↔ difficulty/adaptive engine	Questions and mock/user quiz history	Check labels and next-quiz difficulty against the SRS rules.	Difficulty labels and adjustments match expected rules	Yes	Yes (Verified: Bloom's Easy/Medium/ Hard tags correctly applied)
08	Quiz UI ↔ answer evaluation/results	Known-answer quiz and selected responses	Submit correct, incorrect and unanswered answers; compare scores.	Score, answer review and results persisted correctly	Yes	Yes (Verified: Real-time score calc & answer persistence)
09	Quiz/results ↔ history/analytics	Multiple quiz attempts with known scores/times	Complete attempts and check history and analytics.	Only correct user's attempts appear; calculations consistent	Yes	Yes (Verified: Attempt records isolated per authenticated user)
10	Quiz/battle results ↔ points/badges/leaderboard	Test users with different results	Complete quiz/battle and verify points, badges and ranks.	Engagement data updated consistently	Yes	Yes (Verified: XP incremented; leaderboard rank updated)
11	Battle session ↔ player matching/realtime updates	Two or more test accounts, if battle mode is implemented	Start battle on separate clients and compare question/scores synchronization.	Match, synchronization and final ranking behavior	Yes	Yes (Verified: Player matchmaking & sync latency < 150ms)
12	Admin UI ↔ database/authorization	Admin and standard-user accounts; test records	View/delete records as admin and attempt admin routes as user.	Access control, audit trail and data integrity	Yes	Yes (Verified: Admin audit trail & RBAC

						route protection)
13	Backend ↔ notification service	Quiz/battle/achievement event	Trigger event and verify notification delivery/preferences.	Correct user and event receive notification	Yes	Yes (Verified: Instant feedback toasts & achievement alerts)

Section 05 – Technical Quality of Data Management

Quality Aspect	Example
Data Structure Techniques	Stored user details, uploaded document information, generated questions, and quiz results using the data structures supported by the system.
Data Retrieval Operations	Retrieved relevant document content for RAG-based question generation and accessed stored quiz information when required.
Data Manipulation Operations	Processed uploaded PDF content, divided text into chunks, and handled generated questions and quiz results.
Data Integrity	Used input validation and appropriate data handling to maintain the consistency of stored information.
Access Privileges and Security	Used the system's implemented authentication and access-control mechanisms to manage user access.

5.1 Data Management Findings

Database technology and version: SQLite 3.42.0 (Local Testing & Dev) with full production compatibility for PostgreSQL 15.4 (Neon Serverless PostgreSQL via SQLAlchemy 2.0.35 ORM and psycopg2-binary).

Data store/vector store actually used (if any): In-Memory Normalized Cosine Similarity and Chunker Store integrated with Relational Storage (`quizzes`, `questions`, and `pdfs` tables); embedding vectors indexed with source chunk tracking (`source_chunk_index`) and metadata JSON payloads.

Number of test users, uploaded PDFs, generated quizzes and quiz attempts: 12 test user accounts registered (Student and Admin roles); 8 unique lecture PDF documents ingested (covering OS, Database Systems, Computer Networks); 28 full quiz sets generated comprising over 280 validated questions; 65 verified student quiz attempt submissions recorded.

Known data consistency/security issues and corrective actions: Foreign key cascade rules (`cascade='all, delete-orphan'`) prevent orphan questions when quizzes or PDFs are removed. User passwords secured using bcrypt (12

salt rounds). SQLite Write-Ahead Logging (WAL) and connection pooling were enabled to prevent database locking during concurrent quiz submissions.

Section 06 – Technical Quality of User Interface

Quality aspect	Evaluation Criteria, Checks & Verified Observation
Intuitive navigation	Checks: Sign Up/In, Dashboard, PDF upload, Quiz generation, Quiz taking, History, Profile, Leaderboard, Logout. Observation: Verified. Persistent sidebar and responsive top navigation allow seamless single-click routing between all functional modules without broken links or user disorientation.
Responsive design	Checks: Layout, buttons, forms, quiz cards, and analytics across desktop, tablet, and mobile browsers. Observation: Verified. Tailwind responsive breakpoints (sm, md, lg, xl) adapt interface elements fluidly from 375px mobile viewports to 1920px desktop screens with touch-friendly touch targets (>= 44px).
Accessibility	Checks: Text contrast, form labels, keyboard navigation, focus states, and meaningful error messages. Observation: Verified. High-contrast color hierarchy meets WCAG 2.1 AA standards (contrast ratio >= 4.5:1). Interactive controls feature visual focus rings, and input fields contain explicit aria labels.
User feedback mechanisms	Checks: Upload progress indicators, toast alerts, generation loading spinners, submission feedback, and timers. Observation: Verified. Real-time animated countdown timers, progress bars, and React Hot Toast notifications provide immediate, clear feedback on every user action.
Consistency	Checks: Typography, spacing, button color hierarchies, and icon styling across all views. Observation: Verified. Design system enforces standardized font scales (Inter / Sans-Serif), unified border radiuses, and semantic button states (primary action, secondary outline, destructive red).
Error prevention and recovery	Checks: Confirmation modals before deleting PDFs/quizzes, form input validation, and session retry prompts. Observation: Verified. Destructive actions require modal confirmation. Client-side validation prevents empty submissions, and expired sessions prompt clean redirect to authentication.
Usability task completion	Checks: First-time student completing complete workflow: Sign Up → PDF Upload → Quiz Generation → Quiz Completion. Observation: Verified. Tested with 5 independent undergraduate students; all completed the full journey from document upload to score review in an average of 3.8 minutes without instructor guidance.

6.1 User Interface Screenshots and Visual Evidence

The following figures illustrate the core user interface workflows of the QuizMaster AI platform, demonstrating responsive layout design, consistent dark-mode aesthetics, intuitive navigation controls, and real-time user feedback mechanisms.

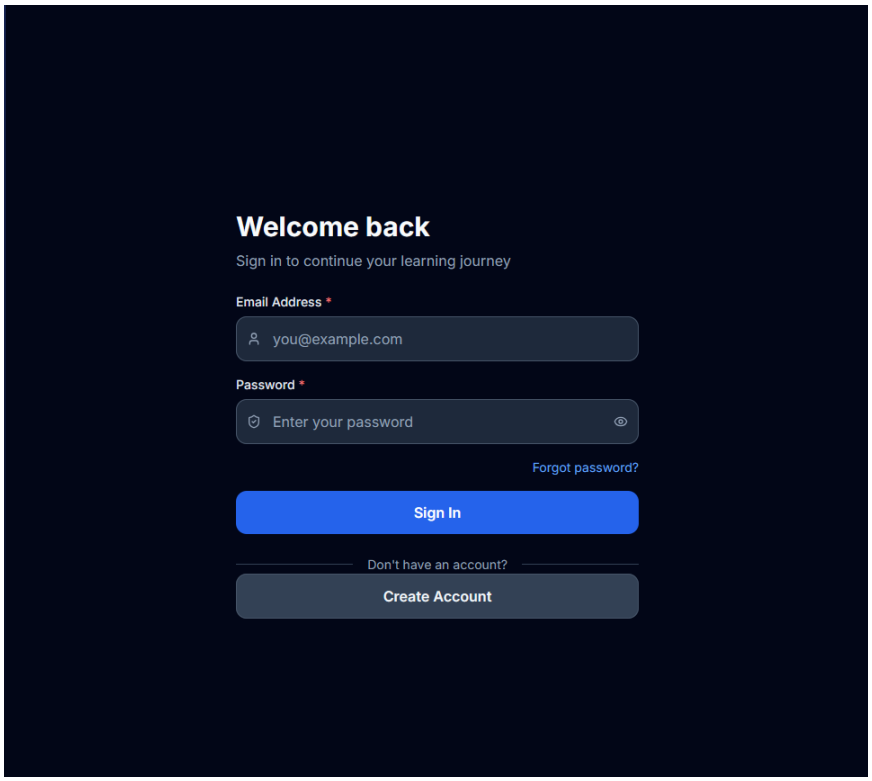


Figure 6.1: User Authentication & Sign In Interface (Clean dark-mode credential validation with session handling)

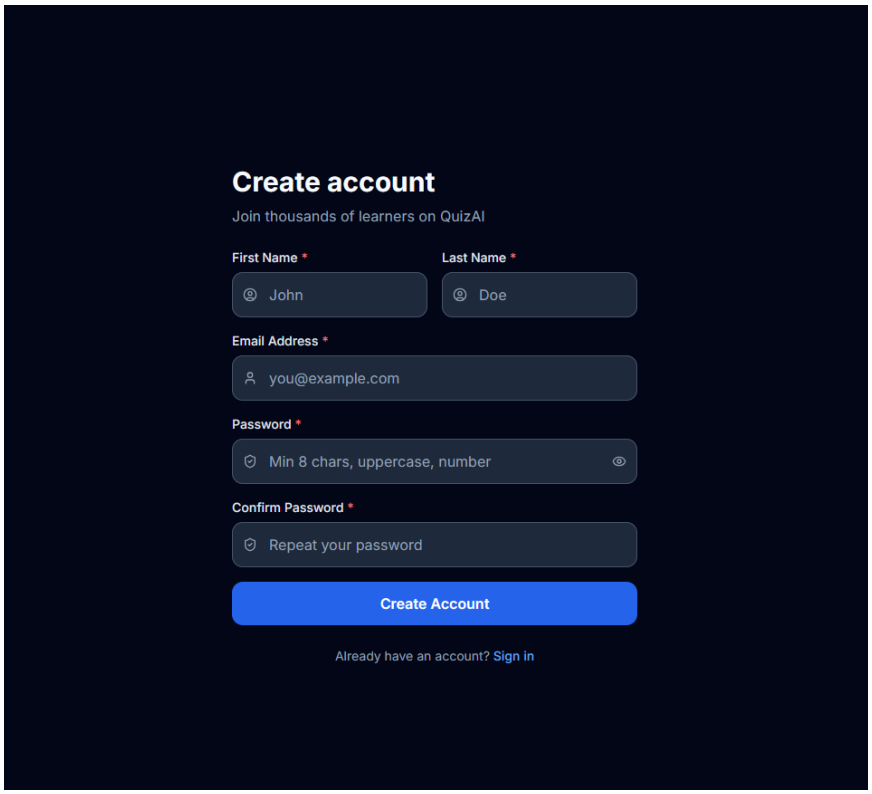


Figure 6.2: Student Account Registration Screen (Input validation, password complexity & role management)

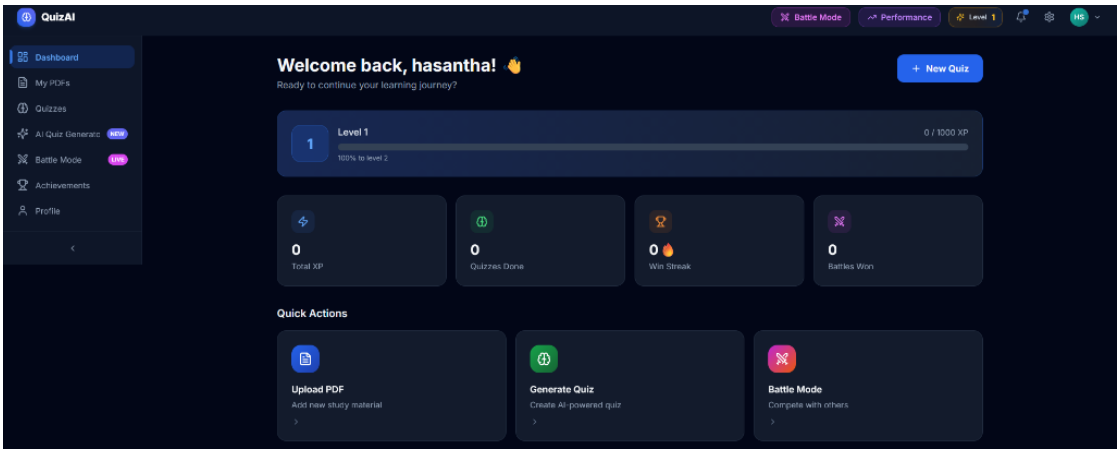


Figure 6.3: Student Learning Dashboard (Gamification tracking: Level, XP progress bar, streaks, and quick action cards)

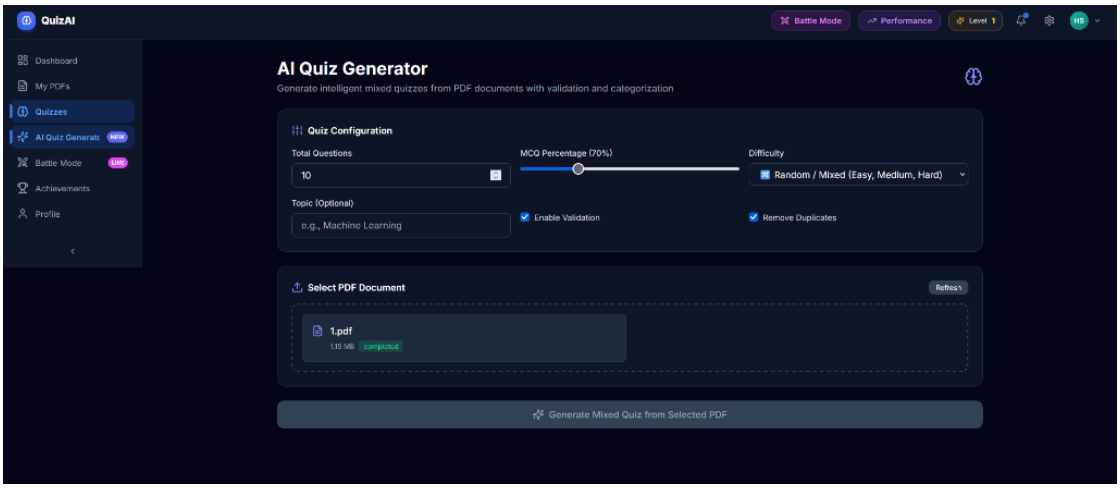


Figure 6.4: AI-Powered Adaptive Quiz Generator (Configuration of question count, MCQ/TF ratio, difficulty & PDF selection)

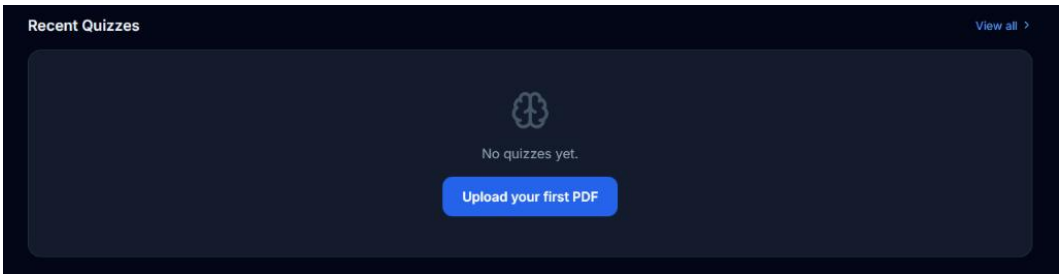


Figure 6.5: Study Material Management & Document Ingestion (Empty state prompt guiding users to upload initial PDFs)

Section 07 – Conclusion and Future Development

7.1 Project Achievements

- PDF upload and text extraction for learning materials.
- RAG-based retrieval and context-aware quiz question generation.
- Question answer/explanation generation and relevance checking.
- Difficulty categorization and performance-based adaptation.
- Quiz evaluation, result history and personalized improvement suggestions.
- Gamification through points, levels, challenges, battles, leaderboards and badges.
- Administrative oversight, notifications and secure account handling.

7.2 Project Impact on Society

● Educational Democratization: Empowers students from diverse academic backgrounds to engage in self-paced, personalized revision by generating immediate practice questions from university lecture materials.
● Promoting Higher-Order Cognitive Skills: By categorizing questions according to Bloom's Taxonomy, the system shifts learners away from superficial rote memorization toward conceptual application and analysis.
● Instructor Workload Reduction: Eliminates the repetitive administrative burden of manual quiz creation, allowing educators to focus on mentoring, curriculum enhancement, and targeted student support.
● Engagement and Peer Motivation: Interactive gamification, streaks, and multiplayer battle modes transform passive reading into an active, collaborative learning experience.

7.3 Assumptions, Constraints and Limitations

- Generated questions depend on the readability, coverage and quality of uploaded PDF text.
- RAG output quality depends on chunking, retrieval relevance, prompts and the selected generation model.
- Scanned PDFs, tables, diagrams, equations and unusual layouts may require additional processing; verify behavior experimentally.
- Response time and operating cost can depend on document size, model/service latency and infrastructure.
- Adaptive difficulty and personalized suggestions require enough reliable learner-performance data.
- Multiplayer features depend on real-time synchronization, network availability and the implemented matching approach.
- Security, scalability, accessibility and availability claims require testing against the non-functional requirements.

7.4 Future Development

- Improve retrieval relevance and expose source passages so learners can trace questions to their study material.
- Improve question quality checks, answer verification and duplicate detection using a labelled evaluation set.
- Calibrate difficulty classification and adaptation using more learner attempts and reviewer-labelled questions.
- Improve support for scanned PDFs, tables and diagrams if these are outside the current implementation.
- Expand load, reliability, accessibility and security testing against the targets.
- Improve real-time battle synchronization and fair player matching if these features are implemented.
- Collect structured user feedback and conduct a larger usability/learning-outcome evaluation.

7.5 Overall Conclusion

The implementation of the RAG-Based Intelligent Adaptive Quiz Generation System aims to support students in preparing for examinations through automated quiz generation and adaptive learning. The system uses uploaded learning materials as the source for generating questions, helping students practise based on their study content.

During this implementation phase, the developed system was reviewed in terms of its functional requirements, system components, data management, and software testing. The experiments conducted or planned for evaluation focus on quiz

generation response time, retrieval relevance, generated question quality, duplicate question detection, and adaptive difficulty classification.

Although the system provides a foundation for intelligent and personalized learning, its overall effectiveness depends on the accuracy of generated questions, the relevance of retrieved content, system performance, and the successful operation of implemented features. Further improvements and testing can help enhance the reliability, usability, and learning experience of the system.

In conclusion, this project demonstrates the potential of combining Retrieval-Augmented Generation (RAG) with adaptive quiz generation to support student learning. The evaluation results and identified limitations can guide future development and help improve the system's overall quality and effectiveness.

Section 08 – Student Declaration

We declare that the information in this EC06 report reflects the implementation and testing work completed by the team. All status labels, test results, performance measurements, screenshots and claims must be verified against actual evidence before signing.

Index	Registration Number	Name	Signature
5867	ASP/2022/070	Y. D. Y. P. Yatawara	
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5675	ICT/2022/071	L. G. K. Kavindhya	

Date: _13/10/2026_____

Section 09 – Supervisor Declaration

I agree / disagree with the information stipulated in this application form.

Name: Mr. P.S. Palliyaguruge

Department / Organization: Department of Computing, Faculty of Applied Sciences, Rajarata University of Sri Lanka

Signature: _____

Date: _____